

Abstract

Current music style transfer research frequently relies on superficial audio-to-audio mapping, resulting in transformations that feel like a digital coat of paint [7]. This proposal shifts toward **Deep Generative Genre Remastering**, a framework designed to deconstruct a song into its architectural components and rebuild it from the ground up. By utilizing disentangled latent representations, we separate invariant musical content from genre-specific stylistic markers [2, 5]. Our approach leverages a **Neural Codec-based Translator** and a **MERT-based stylistic microscope** to synthesize high-fidelity audio that sounds as if it were originally conceived within the target genre [19, 21].

Introduction

Most people treat style transfer as a simple signal manipulation problem, but we are moving away from that superficial approach toward a method that treats a song like a piece of architecture. Instead of just layering a new texture over the original audio, we want to deconstruct a song to its foundation so we can rebuild it properly. We want to separate the content; specifically the melody, harmony, and rhythm, from the style, including instrumentation, tempo, articulation, and arrangement. By remastering the track from the ground up, we ensure the final product is a complete structural overhaul rather than just a filtered version of the source [7].

Brief Summary of Existing Work

The progression of music style transfer research can be categorized into three distinct movements: the initial focus on superficial timbre mapping, the emergence of latent disentanglement techniques, and the recent shift toward high-fidelity generative synthesis. In Section 0.1, we discuss the foundational theories and the limitations of early "filter-based" models. Section 0.2 covers the methodologies for musical deconstruction and disentanglement, while Section 0.3 explores the latest generative architectures capable of structural reconstruction.

0.1 Foundational Theories and the Timbre Ceiling

Traditionally, music style transfer has been treated as a problem of signal manipulation rather than architectural reconstruction. Early position papers, most notably Dai et al. [7], explicitly define the "coat of paint" problem by distinguishing between true style transfer and simple timbre transfer. They argue that a majority of existing work merely changes the "texture" of the sound, the surface-level acoustic characteristics, without touching the underlying composition [7]. This often leads to a "timbre ceiling" where, for instance, a piano piece converted to a violin style sounds like a piano being played through a violin-shaped filter rather than a composition written for strings. Furthermore, the reliance on vanilla end-to-end approaches often yields poor, noisy results because the models attempt to match statistical patterns in raw audio rather than understanding the hierarchical nature of music [7].

0.2 Deconstruction via Latent Disentanglement

To move beyond superficial filtering, research shifted toward deconstructing music into its constituent parts. Hung et al. [2] pioneered methods to separate the "composition", the sequence of notes and timing, from the "timbre" or style of the performance. By creating a latent space where these features are independent, they demonstrated that it is possible to swap the "skin" of a song while preserving its "heart" [2]. This is technically achieved through disentangled representations, where specific encoders are built to isolate pitch from stylistic markers [2].

Furthering this concept of the "musical skeleton," Yang et al. [5] addressed the problem of disentanglement by pulling audio apart into three independent factors: pitch, rhythm, and chords. Their work utilizes explicitly constrained latent spaces, ensuring that stylistic artifacts, such as the swing of a jazz drum kit or the distortion of a rock guitar, do not bleed into the core melodic code [5]. This provides the scientific validation for our "bricks" metaphor; by training an encoder to map these features to different parts of a latent space, we can hold the pitch steady while completely replacing the rhythmic or chordal structure [5].

0.3 Genre Recognition and the Target Blueprint

While deconstruction isolates the source, defining the destination requires a sophisticated understanding of genre. Ghosal and Kolekar [6] moved beyond simple labeling by using an ensemble of CNN-LSTM models and transfer learning to extract deep stylistic features. Their work demonstrates that a genre is best represented not as a flat category, but as a rich, 160-dimensional "target vector" that encapsulates mood, instrumentation, and era [6]. By analyzing spectral and rhythmic features like Mel-Spectrograms, Tonnetz (harmonic centroids), and Tempograms, they provide the necessary "architectural blueprint" that a generative system needs to rebuild a song from the ground up [6].

0.4 Generative Synthesis and Architectural Reconstruction

The final phase of research involves the actual synthesis of high-fidelity audio from these abstract representations. Engel et al. [4] introduced GANSynth, proving that Generative Adversarial Networks (GANs) could produce coherent audio much faster than autoregressive models by focusing on the relationship between magnitude and frequency in the spectral domain. This suggests that a decoder can act as an architect, understanding global structure rather than just local samples [4]. While early attempts like WaveGAN [9] demonstrated that GANs could achieve global coherence in one-second raw waveform slices, they often struggled with high-resolution stability. We build on this adversarial foundation but pivot to codec-latent translation to avoid the temporal resolution hurdles identified in their original work [9, 19]. In parallel, recent advances in Diffusion Models [1] have shown that musical attributes can be captured and transferred with minimal data, allowing for the precise stylistic reconstruction of specific instruments [1].

However, maintaining structural coherence in long-form music remains a challenge. Roberts et al. [3] proposed a hierarchical latent vector model (MusicVAE) to manage long-term structure. By organizing the latent space into a hierarchy, their model acts like a musical conductor,

ensuring the decoder stays faithful to the original melodic intent while managing complex arrangements [3]. Finally, the StyleFormer architecture [8] suggests that transformer-driven composition can ensure that style is applied with full consideration of the semantic content, building a bank of style patterns to selectively reconstruct the output [8]. Together, these works provide the technical foundation for a system that doesn't just paint a house, but rebuilds it brick-by-brick into a new musical mansion.

Our Implementation Plan

Our technical strategy centers on building a robust pipeline that prioritizes musical disentanglement [2, 5]. The first stage involves the development of the Deconstruction Encoder to strip away stylistic markers and isolate the essential musical DNA. Simultaneously, we integrate a dedicated Genre Extraction module [6, 21]. By utilizing a pre-trained neural network like a Transformer-based classifier, we can articulate the defining features of a target genre into a target vector. The final stage is the Reconstruction Decoder, where the actual generative remastering occurs using either a GAN [20] or a Neural Codec [19].

What You Plan To Do

Our research focuses on transcending the "digital coat of paint" limitation [7] through five sequential development phases.

- Lab 1: Establishing the Musical Skeleton (Deconstruction Encoder): The primary objective is to develop a Deconstruction Encoder that extracts a genre-agnostic representation. Using the principles of latent representation disentanglement [5, 2], we will separate invariant musical content from stylistic noise.
- Lab 2: Stylistic Intelligence and Target Vector Calibration (Genre Extraction): We will train an ensemble of deep neural networks [6] to identify defining features of musical universes. This focuses on extracting a 160-dimensional target vector that serves as the architectural blueprint.
- Lab 3: Architectural Reconstruction and Generative Synthesis (Reconstruction Decoder): In this phase, we implement the Reconstruction Decoder using Neural Codec Latents [19] or GANs [4, 20]. The decoder synthesizes entirely new audio conditioned on the skeleton and the target vector.
- Lab 4: Structural Coherence and Long-Form Remastering: This lab utilizes hierarchical latent vectors [3] and semantic-acoustic splitting [22] to manage long-term structural consistency.
- Lab 5: Final Evaluation and Qualitative Validation: The final phase validates the outputs through blind listening tests and acoustic realism metrics like Fréchet Audio Distance [20].

Timeline:

Week 1-2: Reviewing sources, understanding problem scope

Week 2: Starting on lab 1 and testing out the theory of our proposal

Week 3: Start working on lab 2 and ensure our basis for the generation labs is solid

Week 4: Start working on lab 3, attempt to train a model that is able to balance all of our desired metrics and goals

Week 5: Continue working on our generation step, ensuring that the 5 second ‘bricks’ we generate do not have artifacts that will compound once we work on cohesion, start lab 4

Week 6: Finish lab 4, working on maintaining cohesion long term and ensuring our generations don't let the error compound out, start working on lab 5 preparing generated samples to decide on the merits of our approach vs traditional means.

Week 7: Finish lab 5 to compare metrics, gather our findings and clean up project for presentation

Week 8: Finalize our presentation, practice our presentation and finalize anything needed for our demos.

Description of lab 1

For Lab 1, we are going to build an adversarial deconstruction encoder to pull out a musical skeleton that stays consistent regardless of genre. The goal is to separate audio into two different latent spaces: one for content (Z_c) and one for style (Z_s) [2, 5]. To make sure these are actually separate, we plan to use a gradient reversal layer [2]. This adversarial setup forces the content head to hide any stylistic info from a discriminator, which effectively strips away things like instrumentation or swing from the melodic bricks [7].

We also want to use a teacher-student anchor to keep the long-term structure stable. This works by using a distillation loss to stop the melodic representation from drifting away from a pre-calibrated teacher model. We will verify the success of this deconstruction using a disentanglement audit. Our data pipeline will pull from several different sources to provide the right building blocks. We are integrating hip hop drum loops for rhythmic DNA [12, 13] and classical scores to help the encoder map out structural blueprints like articulation and dynamics [14, 16]. We also plan to use a 40GB repository of raw audio for high quality textures [15] and pull isolated instrument samples from public timbre libraries [18]. To keep the model from hallucinating, we will use speech data from the LibriSpeech ASR corpus [10] as a hard negative class to help the encoder isolate actual music from vocal noise.

The lab is considered complete only when the model passes four strictly defined scientific thresholds:

- Content Purity: (Content Leakage ≤ 0.15)
- Style Intelligence: (Style Accuracy ≥ 0.85)
- Music Gate Resolution: (Gate AUC ≥ 0.90)
- Melodic Invariance: (Cosine Similarity ≥ 0.92 across dual-soundfont renders)

By hitting these targets, we ensure that our "universal skeleton" is a high-fidelity, style-neutral representation capable of driving the generative reconstruction in Lab 3.

Description of lab 2

With the deconstruction encoder frozen, Lab 2 focuses on creating the blueprints for the new mansion by defining a genre target vector space. We are building a 160-dimensional coordinate system that captures the actual soul of a genre [6, 21]. We will do this by harvesting stylistic markers from thousands of tracks across our SoundSafari and Hip Hop datasets [15, 12]. The encoder will extract the style latents (Z_s) and combine them with melodic descriptors to create a rich database of musical textures.

The primary goal is to calculate genre centroids, which are the mathematically averaged coordinates for each musical style. To make sure these blueprints are distinct, we will use linear discriminant analysis to project these 160-dimensional vectors into a lower-dimensional space. This allows us to perform a centroid validation audit, using clustering algorithms to prove that genres like Baroque and Boom-Bap are well-separated and mathematically unique.

We will also use linear probes to verify that these target vectors actually contain enough information to guide the reconstruction process. Success in this lab depends on reaching a silhouette score of at least 0.45 and ensuring that our inter-centroid separation remains stable across different samples. By the end of this lab, we should have a precise, noise-free blueprint that tells the generative engine in Lab 3 exactly what instrumental and textural palette to use for the remaster.

Description of lab 3

In this lab, we plan to move away from raw spectrogram generation, which historically led to unstable "screeching" and robotic artifacts, and instead build a Reconstruction Decoder based on neural codec latents[19]. We plan to utilize the Facebook EnCodec architecture as a high-fidelity foundation. By operating directly on the codec's quantized embeddings, we'll use the pretrained decoder as a quality guardrail[19]. Hypothetically this ensured that our output always remains in the realm of "real audio," allowing the model to focus entirely on the style transformation rather than learning the basic physics of sound from scratch.

The core of our implementation is a Codec Latent Translator that uses FiLM (Feature-wise Linear Modulation) to inject style. Unlike a traditional autoencoder, our translator is conditioned on two distinct signals: the genre-agnostic "universal skeleton" (Z_{content}) from Lab 1 and a high-dimensional MERT-based style vector (Z_{style}) which acts as our stylistic blueprint[21].

We are also proposing a Direct Output Architecture for this phase. By removing the "residual" constraint that usually forces models to stay too close to the source audio, we intend to allow the translator to fundamentally rewrite the latent code[23]. This should result in a much more convincing genre shift than standard additive methods.

To ensure the output sounds like the target genre without losing the original melody, we will utilize a multi-loss training stack. This includes a MERT-based Style Probe to drive the stylistic transformation and a Multi-Scale Waveform Discriminator using a hinge-loss GAN objective to maintain acoustic realism[20]. Success will be measured by hitting a style confidence and accuracy of ≥ 0.85 while maintaining a melodic precision score of ≥ 0.90 .

Description of lab 4

While the short-form synthesis in our earlier labs handled five-second segments effectively, moving to full-length tracks introduces the risk of losing long-range musical structure and unity. To solve this, Lab 4 focuses on maintaining internal consistency across several minutes of audio to prevent structural drift. We will implement a conducting layer using hierarchical latent vectors[3], which provide a structured global outline for the decoder to follow.

Instead of treating each audio chunk as an independent snippet, we will use a seamless mosaic assembly that relies on overlapping latent windows and trajectory interpolation[3, 22]. This smoothing process ensures that transitions between consecutive style vectors remain fluid, preserving the phrasing and rhythmic thread of the original song. By anchoring the generation to this hierarchical map, we reduce compounding errors and ensure the final remaster feels like a cohesive performance rather than a series of disconnected bricks. We are going to evaluate our results by looking at both the hard data and how humans actually perceive the final music. We want to see if listeners can actually tell that our version sounds like a full arrangement rather than just a digital effect.

Description of lab 5

The final phase of our research is the perceptual audit, where we move past raw numbers to see if our remastered tracks actually hold up to human ears. While metrics like style confidence and melodic precision give us a mathematical baseline (Style Conf ≥ 0.85 , MPS ≥ 0.90), the ultimate goal is for the music to sound like it was originally conceived in the target genre. We are answering two fundamental questions: does the output actually embody the new genre, and is the original melody still clearly recognizable? To prove this, we are setting up blind A/B testing where listeners will compare our architecture-driven results against the traditional "filter-based" style transfer methods. This comparison is vital because it highlights the difference between a superficial timbral layer and a complete structural rebuild.

We are grounding this human feedback with acoustic realism metrics like Fréchet Audio Distance[20], which measures the statistical distance between our generations and a massive set of real-world recordings. This ensures that we aren't just hitting high classification scores

through generative noise but are actually producing high-fidelity textures. We are also tracking subjective genre authenticity to see if a listener can correctly identify the "mansion" we built as belonging to the intended musical universe. By combining these rigorous objective proxies with qualitative blind tests, we can scientifically validate whether our built-from-scratch approach delivers a more convincing and musically coherent experience than simple audio manipulation[7].

[1] [Music Style Transfer with Time-Varying Inversion of Diffusion Models](#)

- Outlines approach to capture musical attributes using minimal data
- Can transfer the style of specific instruments

[2] [\[1905.13567\] Musical Composition Style Transfer via Disentangled Timbre Representations](#)

- The researchers developed a method to disentangle the timbre of the instruments from the actual musical composition, which they define as the sequence of notes and their timing. This is the scientific validation for the deconstruction phase because it proves that you can isolate the specific texture of a rock guitar or a classical violin as a separate variable from the melody itself.
- The authors show that by creating a latent space where timbre and composition are independent, you can effectively swap the skin of a song(surface details) without touching its heart(deeper features).
- The paper highlights that when you successfully disentangle these elements, the resulting transfer is far more convincing because the model isn't trying to force one instrument to sound like another through a filter. Instead, it is using the learned composition to drive a completely different timbre representation.
- It suggests that once the skeleton is isolated, the decoder has the freedom to apply new instrumentation and articulation that are native to the target genre.

[3] [A Hierarchical Latent Vector Model for Learning Long-Term Structure in Music](#)

- The authors of this paper demonstrate the process of using a hierarchical decoder to reconstruct music from a latent
- It shows that by organizing the latent space into a hierarchy, the model can manage complex arrangements and instrumentation while staying faithful to the original melodic and harmonic intent.

[4] [GANSynth: Adversarial Neural Audio Synthesis](#)

- They proved that GANs could generate high-fidelity audio that is much more coherent than older autoregressive models by focusing on the relationship between magnitude and frequency in the spectral domain.
- Proved that a decoder can generate audio thousands of times faster and more consistently than traditional methods if it understands the global structure of the sound rather than just the local samples.

[5] [Deep Music Analogy Via Latent Representation Disentanglement](#)

- This paper addresses the problem of musical disentanglement head-on by proving that you can take a piece of music and pull it apart into three independent factors: pitch, rhythm, and chords. The researchers used what they call explicitly constrained latent spaces to make sure that these elements don't bleed into each other.
- This paper shows that to find the skeleton of a song you can train an encoder to map features (the swing of a jazz drum kit or the distortion of a rock guitar) to different parts of a latent space. This means you can hold the pitch steady while you completely swap out the rhythm or chord structure.

[6][Music Genre Recognition using Deep Neural Networks and Transfer Learning](#)

- While the previous papers focused on how to strip the song down to its skeleton, this work focuses on how to accurately define the destination
- The researchers used an ensemble of CNN-LSTM models and transfer learning to achieve state-of-the-art results
- They explore the breakdown of spectral and rhythmic features like MelSpectrograms, Tonnetz, and Tempogram to show that these specific features are the most effective at capturing the "soul" of a genre.
- By using a transfer learning model originally trained on large-scale music tagging, the authors demonstrate that you can extract a rich 160-dimensional vector that encapsulates more than just a genre name, it captures the mood, instrumentation, and era- which is our target vector

[7][\[1803.06841\] Music Style Transfer: A Position Paper](#)

- It explicitly defines the problem that has been described as a digital coat of paint by distinguishing between style transfer and simple timbre transfer. The authors argue that a lot of what people call style transfer is actually just changing the texture of the sound, which aligns perfectly with the idea that these models are just filtering the audio rather than understanding it.
- The researchers point out that in order for a transformation to be successful, the model must be able to preserve the content, which they define as the pitch and rhythm, while completely replacing the style. This is the scientific foundation for the skeleton metaphor.
- The paper advocates for a more holistic approach that considers the composition and the arrangement, rather than just the raw audio samples and it addresses the ceiling that many researchers hit when they try to use vanilla end-to-end models,

[8][StyleFormer: Real-Time Arbitrary Style Transfer via Parametric Style Composition](#)

- Instead of a one-size-fits-all transformation, the authors propose building a bank of compact style patterns that can be selectively used to reconstruct the output.
- The authors demonstrate that by using parametric modulation, they can create results that are both flexible and faithful to the original structure.
- Although it is focused on images, its core innovation lies in using transformer-driven style composition to ensure that the style is applied with full consideration of the underlying content structure.

[9][Adversarial Audio Synthesis](#)

- The authors introduced WaveGAN to handle raw waveforms and SpecGAN to handle spectrograms. It's highly relevant to our project because they were the first to run into the "phase" issues that cause the robotic screeching we saw in our early Lab 3 runs. By comparing raw audio to image-like representations, they proved that while GANs can learn global coherence in short slices, they struggle with the massive temporal resolution of audio. We use this as the scientific justification for why we moved past raw spectrograms and adopted the Neural Codec path to act as our structural guardrail.

[10][LibriSpeech ASR corpus](#)

- We are utilizing this massive collection of clean speech as a Non-Musical Contrastive Baseline. By processing non-musical audio through the same pipeline, we train the Deconstruction Encoder to distinguish between "human performance noise" and actual "melodic bricks." This ensures the isolated skeleton is purely musical and not contaminated by vocal prosody or acoustic artifacts.

[11][TheSession-data](#)

- This is our primary source for Rhythmic and Melodic Invariance. TheSession provides thousands of traditional tunes in ABC notation, the ultimate genre-agnostic representation. We use this to train the model on how a core melody (the "tune") remains identical even when the "paint" (instrumentation or regional style) changes completely.

[12][XTc Files of Hip Hop](#)

- A specialized archive for Genre-Specific Rhythmic DNA. These files provide the raw breakbeats and percussive textures necessary for the Reconstruction Decoder to understand the "swing" and "groove" of hip-hop. This prevents the model from sounding like a sterile MIDI sequencer when building a track in this universe.

[13][WaivOps-HH-LFBB](#)

- This corpus of Lo-Fi and Boom-Bap loops serves as the Micro-Timing Reference. We use these loops to teach the model the architectural difference between a perfectly quantized beat and the "humanized" displacement that defines modern urban genres, ensuring our generative remastering captures the "soul" of the target style.
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[14][petal_bach_goldberg-variations](#)

- Our Classical Anchor Corpus. By using these high-quality, technically demanding variations, we provide the model with a "gold standard" for complex harmonic structures. This is the benchmark for our "rebuilding" phase, if the model can convincingly remaster a Bach variation, it can handle the architectural complexity of any target genre.

[15][CC0-1.0-Music](#)

- This 40GB collection provides the sheer acoustic diversity needed to train a robust Genre Extraction module. It ensures our 160-dimensional target vector is calibrated against a wide variety of real-world "mansions," from experimental electronic to folk.

[16][PDMX](#)

- The Architectural Blueprint Teacher. Unlike MIDI, MusicXML preserves articulations, dynamics, and phrasing. We use PDMX to guide the Reconstruction Decoder in how to "play" the bricks, telling it not just which notes to hit, but the specific intensity and style required for a realistic performance.

[17][Lakh MIDI Dataset Clean](#)

- Acting as our Symbolic Rosetta Stone. We utilize the "Clean" subset specifically to avoid the "corrupted brick" problem of the larger LMD. This dataset provides the ground-truth skeletons for tens of thousands of songs, allowing us to perform supervised disentanglement by matching known MIDI structures to their audio counterparts.

[18][freesound-datasets Public](#)

- The Timbre Component Library. We pull isolated instrument samples from Freesound to populate our "Timbre Latent Space." This allows the Reconstruction Decoder to select the specific "building materials", a certain violin texture or a specific synth pluck, needed to rebuild the song with high-fidelity realism.

[19] [High Fidelity Neural Audio Compression](#)

- This is the foundation for our Lab 3 pipeline. These researchers built the neural codec we're using as our quality guardrail. It's why our audio sounds professional even before we start the style transfer. It proves that working with quantized "codebook" latents is way more stable than trying to generate raw audio samples from scratch, because the codec already understands the basic laws of sound.

[20] [HiFi-GAN: Generative Adversarial Networks for Efficient and High Fidelity Speech Synthesis](#)

- Proved that if you use a Multi-Period Discriminator, you can actually capture the tiny rhythmic patterns in audio that make it sound "real" instead of digital. We're using their logic in our reconstruction decoder to make sure our "mansion" has a realistic acoustic texture instead of just sounding like a fuzzy spectrogram.

[21] [MERT: Acoustic Music Understanding Model with Large-Scale Self-supervised Training](#)

- MERT is a massive transformer model trained specifically to "understand" music, and we're using it as our stylistic microscope. It gives us those high-dimensional vectors that act as our target blueprint. It's way better than a standard classifier because it can see deep features like instrumentation and era that other models miss.

[22] [AudioLM: a Language Modeling Approach to Audio Generation](#)

- This paper validates our "Semantic + Acoustic" split. They proved that you can keep a song's long-term structure consistent by separating "what" is being played from "how" it sounds. It gives us the technical backing for our Lab 4 plan, showing that we can use these high-level tokens to keep our 3-minute tracks from drifting or becoming a mess of random notes.

[23] [BigVGAN: A Universal Neural Vocoder with Large-Scale Training](#)

- These authors showed that if you train on a massive variety of audio, you can create a "universal" decoder that works on any genre without needing a residual "shortcut" to the source. It's the scientific proof that we can safely remove the identity leashes and still get a high-quality result that actually sounds like the target genre